Quantum Fault Tolerance Meets Toom's Contours

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1 Computation Model

Informally, a quantum circuit is a placement of a collection of quantum gates, taken from a finite set, in different space-time locations. Each space-time location, associated with a gate, encodes the time of the application and the qubits in the support of the gate. Formally

Separate between an element gate $g \in \mathcal{G}$ and a gate, which also has a space-time location.

Definition 1.1 (Quantum Circuit)**.** A space-time location l is a tuple that encodes in its first coordinate time, and in its second coordinate a subset of integers referring to a support of (q)bits. For a collection of quantum gates $\mathcal{G} = \{g \colon L(\mathcal{H}_2^{\otimes \Delta}) \to L(\mathcal{H}_2^{\otimes \Delta})\}$, and integers $w, d \in \mathbb{N}$ a quantum circuit C, with depth d and width w, is a collection of tuples $\{u_i = (g_i, l_i)\}_{i=1}$, where u_i are the gates of C and each u_i is composed of a gate $g_i \in \mathcal{G}$ and a space-time location $l_i \in \mathbb{Z}_d \times \mathbb{Z}_w^{\Delta}$. The gates of C have to satisfy the constraint that for every two gates with the same time coordinate, there cannot be a non-empty intersection in their space coordinate, namely for $u_i = (g_i, l_i), u_j = (g_j, l_j) \in C$, the equality $l_{i,1} = l_{j,1}$ implies $l_{i,2} \cap l_{j,2} = \emptyset$. We define the t-cut of C to be $C|_t := \{u_i = (g_i, l_i) \colon u_i \in C \text{ and } l_{i,1} = t\}$.

To define the running of a quantum circuit C on a given (mixed) quantum state, we will need to have the notion of the directed acyclic graph associated with C.

Definition 1.2 (Associated Computation DAG)**.** Let $C = \{u_i\}_{i=1}$ be a quantum circuit with depth d and width w. We define the associated computation DAG of C, denoted by π_C, to be the acyclic directed graph $\pi_C = (\mathcal{U}, E)$ as follows:

1. The vertex set $\mathcal{U}$ is the gate set of C, namely $\mathcal{U} = \{u_i\}_{i=1}$.
2. For $u_i = (g_i, l_i), u_j = (g_j, l_j)$, There is an edge $\{u_i, u_j\} \in E$ if and only if:

$$j \in \arg\min_k \left\{ l_{k,1} : (g_k, l_k) \in \mathcal{U}, l_{k,1} > l_{i,1} \text{ and } l_{k,2} \cap l_{i,2} \neq \emptyset \right\}$$

   Note that there might be multiple indices that get the minimum, so the out degree might be greater than one.

In addition, suppose that $J = \{j_1, j_2, \ldots, j_k\} \subset [|C|]$ is an order of the indices of the gates in C according to the topological order induced by π_C. Then, we say that J is an order of indices that agrees with π_C and define the series of circuit $C_J = \{C_{J,i}\}_{i=1}$ as follows:

1. The first element: $C_{J,1} = \{u_{j_1}\}$
2. Then, $C_{J,i+1} = C_{J,i} \cup \{u_{j_i}\}$. We name $C_{J,i}$ the partial computation circuit until step i.

Definition 1.3. Let ρ be a density matrix over w qubits, and let $C = \{u_i\}_{i=1}$ be a quantum circuit with depth d and width w. For any density matrix σ, quantum gate $g \in \mathcal{G}$, and location l, the application of the gate-location tuple (g, l) on σ, denoted by $(g, l) \circ \sigma$, is the application of g on σ where wires are ordered such that the first qubit in the input to g is $l_{2,1}$, the second is $l_{2,2}$, and so on. The result of the application of C on ρ, denoted by $C \circ \rho$, is given by iterative applications of the gates $\{u_i\}_{i=1}$ on ρ, according to a topological order induced by π_C.

We say that $\rho_1, \rho_2, \ldots, \rho_k$ is a computation prefix if $\rho_1 = \rho$ and $\rho_{i+1} = u'_i \circ \rho_i$, where $\{u'_i\}_{i=1}$ is a prefix of a topological ordering of $\{u_i\}_{i=1}$. If $\{u'_i\}_{i=1}$ contains all the gates in $\{u_i\}_{i=1}$, then we call $\rho_1, \rho_2, \ldots, \rho_k$ a running.

Definition 1.4 (C Subjected to $\mathcal{E}$)**.** Let $C = \{u_i\}_{i=1}$ be a quantum circuit with depth d and width w. For a set of fault locations $\mathcal{E} \subset [d] \times [w]$, we define the quantum circuit C *subjected to* $\mathcal{E}$, denoted by $C[\mathcal{E}] = \{u'_i\}_{i=1}$, to be the quantum circuit that is the result of the following map:

$$u_i = (g_i, (l_{i,1}, l_{i,2})) \mapsto \begin{cases} (g_i, (2l_{i,1}, l_{i,2})) & u_i \in C \\ (g_i, (2l_{i,1}-1, l_{i,2})) & u_i \in \mathcal{E} \end{cases}$$

We will denote the associated computation DAG of C affected by $\mathcal{E}$ as $\pi[C, \mathcal{E}]$.

Definition 1.5 (Faults Propagation to a Step τ)**.** For a quantum circuit $C = \{u_i\}_{i=1}$, with depth d, an order $J = \{j_1, j_2, \ldots, j_k\} \subset [|C|]$ that agrees with π_C, and gate u_i in C, a propagation of a gate $u_i \in C$ to step τ is the quantum operator $X \colon L(\mathcal{H}_2^w) \to L(\mathcal{H}_2^w)$ obtained by the follow process:

1. First, check if i precedes τ in J. If not, then return $X = I$. Otherwise, denote $j_l = i$, and look for an operator X_1 such that $u_{j_l} u_{j_{l+1}} = u_{j_{l+1}} X_1$.
2. Then, for $k \leq \tau - l$, look for an operator $X_k \colon L(\mathcal{H}_2^w) \to L(\mathcal{H}_2^w)$, such that $X_{k-1} u_{j_{l+1}} = u_{j_{l+1}} X_k$.
3. At the final stage of the process, namely where $k = \tau - l$, we define $X := X_k$ to be the propagation of u_j to step τ.

For a quantum circuit C, a set of faults $\mathcal{E} = \{e_i\}_{i=1}$, an order $J \subset [|C[\mathcal{E}]|]$ that agrees with $\pi_{C[\mathcal{E}]}$, and a step $\tau \in J$, denote by $X^\tau_{e_i}$ the propagation of $e_i \in \mathcal{E}$ in $C[\mathcal{E}]$ to step τ. Let $j'_1, j'_2, \ldots, j'_l$ be the indices of the faults in $\mathcal{E}$ such that their corresponding positions in J precede τ, ordered according to the reverse topological order induced by $\pi_{\mathcal{E}}$. The propagated error to step τ is defined by

$$X^{\tau} := \prod_{i} X^{\tau}_{e_{j'_i}}$$

Definition 1.6 (Deviation)**.** For a quantum circuit C, a set of faults $\mathcal{E}$, and step τ, let X^{τ} be the propagated error at step τ. We define the deviation of $C[\mathcal{E}]$ from C at step τ by the set of indices on which X^{τ} acts non-trivially.

Definition 1.7. Using the same notations as in Definition 1.6, for a set of unitary operators $\mathcal{S} = \{s_i \colon \mathcal{H}_2^w \to \mathcal{H}_2^w\}_{i=1}$, we define the propagated error up to stabilizers in $\mathcal{S}$ as $X^{\tau}[S] := \min_{s \in \mathcal{S}}\{w(sX)\}$. Then, we define the deviation of $C[\mathcal{E}]$ from C at step τ up to stabilizers in $\mathcal{S}$ to be the subset of coordinates on which $X^{\tau}[S]$ acts non-trivially.

The following, extends the notion of Tanner graphs to the context of quantum circuits.

Definition 1.8 (Tanner Graph of a Register)**.** For a quantum circuit C, a register R is a subset of qubits in C. The *Tanner graph* at step τ of register R, denoted by $\mathcal{T}_{R,\tau}$, is the graph defined as follows:

1. If there is a code $\mathcal{C}$, such that at turn τ, the qubits in C are an encoding of $\mathcal{C}$, then $\mathcal{T}_{R,\tau}$ is the Tanner graph of $\mathcal{C}$.

2. Otherwise, $\mathcal{T}_{R,\tau}$ is the empty graph.

Where we consider a range of turns in which R is encoded in the same code, and the turn is clear from the context, we will be brief by omitting the turn subscript.

Definition 1.9 (Generalized Tanner Graph of a Circuit)**.** For a quantum circuit C, and registers a and b in C, the *generalized Tanner graph* of the register obtained by taking their union $a \cup b$ is $\mathcal{T}_{a \cup b} := \mathcal{T}_a \cup \mathcal{T}_b$. For a partition of the qubits into registers $\{R_i\}_{i=1}$, the *generalized Tanner graph* of a quantum circuit C, at step τ, according to the partition $\{R_i\}_{i=1}$, is the graph obtained by:

$$\mathcal{T}_{C,\{R_i\}_{i=1},\tau} := \bigcup_{i=1} \mathcal{T}_{R_i,\tau}$$

When the partition is clear, we will brief and use $\mathcal{T}_C$ to define the generalized Tanner graph of C.

Definition 1.10 (Clustered Deviation)**.** For a quantum circuit C, a register R of C, and a set of faults $\mathcal{E}$, let $\mathcal{T}_R$ be the Tanner graph of R, and let $\mathcal{D}_{\tau}$ be the deviation of $C[\mathcal{E}]$ from C on R at step τ. Denote by $\mathcal{T}_R|_{\mathcal{D}_{\tau}}$ the subgraph of $\mathcal{T}_R$ induced by the restriction of the left vertices (bits) to the indices in $\mathcal{D}_{\tau}$. Let us abuse and extend the connection notion to Tanner graphs by defining that two left vertices are connected if and only if they share a right neighbor. We define the deviation clusters at step τ to be the connected components of $\mathcal{T}_R|_{\mathcal{D}_{\tau}}$. The size of each deviation cluster is the number of right vertices it contains.

Claim 1.11. Let C be a quantum circuit composed of gates with bounded fan degree $\Delta \in \mathbb{N}$. In addition, let R be a quantum register encoded by a quantum error correction code $\mathcal{C}$ with Tanner graph $\mathcal{T}$ that has left and right degrees Δ_1 and Δ_2. Fix a set of faults $\mathcal{E}$, and an order J that agrees with $\pi_{C[\mathcal{E}]}$. For any τ, denote by $Y_1^{(\tau)}, Y_2^{(\tau)}, \ldots, Y_l^{(\tau)}$ the deviation clusters at step $\tau \in J$. Assume that at step $\tau_1 \in J$ there is a deviation cluster $Y_i^{(\tau_1)}$ of size $|Y_i^{(\tau_1)}|$. Then there is a deviation cluster at step $\tau_1 - 1$, let's denote it by Z, such $Z \cap Y_i^{(\tau_1)} \neq \emptyset$ and Z is of size at least:

$$|Z| \geq \frac{1}{\Delta\Delta_1\Delta_2}|Y_i^{(\tau_1)}|$$

Furthermore there is a deviation cluster at step $\tau_1 - 1$, let's denote it by Z', such $Z' \cap Y_i^{(\tau_1)} \neq \emptyset$, and Z' is of size at most:

$$|Z'| \leq \Delta\Delta_1\Delta_2|Y_i^{(\tau_1)}|$$

Add: that when it isn't mentioned, we assume that the error at time t can be decomposed into a product $\Rightarrow$ at each step, a fault is supported on at most one (q)bit.

Proof. Denote by $u = (g, l)$ the gate at step τ. If u acts trivially on the (q)bits in $Y_i^{(\tau_1)}$ then they also belong to deviation on register R at step $\tau_1 - 1$, Since they are connected in $\mathcal{T}$ they form a deviation cluster on register R at step $\tau_1 - 1$ and we got the desired. Thus it is left to handle the case in which u acts non-trivaily on (q)bits in $Y_i^{(\tau_1)}$. Since u acts on at most Δ qubits, and each qubit has at most $\Delta_1(\Delta_2 - 1)$ neighbors at $\mathcal{T}$ it follows that u acts non trivialy on at most

$$\Delta + \Delta\Delta_1 (\Delta_2 - 1) \leq \Delta\Delta_1\Delta_2$$

deviation clusters at step $\tau_1 - 1$. Let l be integer no greater than $\Delta\Delta_1\Delta_2$ and let $Y_1, Y_2, \ldots, Y_l$ be the deviation cluster, on register R, at step $\tau_1 - 1$, on which u acts non trivialiy, since $Y_i^{(\tau_1)} \subset \bigcup_j Y_j$. We have that:

$$\begin{aligned}|Y_i^{(\tau_1)}| &\leq \sum_j |Y_j| \leq \Delta\Delta_1\Delta_2 \max_j |Y_j| \\ \Rightarrow \max_j |Y_j| &\geq \frac{1}{\Delta\Delta_1\Delta_2}|Y_i^{(\tau_1)}|\end{aligned} \tag{1}$$

Complete the second inequality. Should consider to erase what have already written about it. To get the second inequality, we use the fact that u is a unitary gate, in particular, u is reversible. Let $C_{J,\tau_1} = u_1, .., u_{\tau_1}$ be the partial computation circuit until step τ_1. Observe that C_{J,τ_1-1} is obtained by stacking $u_{\tau_1}^\dagger$ as a final stage to C_{J,τ_1}, denoted by $u^\dagger \circ C_{J,\tau_1}$. Thus, the deviation from C_{J,τ_1} at step $\tau_1 - 1$ equals the deviation from $u^\dagger \circ C_{J,\tau_1}$ at step $\tau_1 + 1$. By eq. (1), $\square$

Corollary 1.12 (Intermediate Value Theorem)**.** Let $\alpha = 1/(\Delta\Delta_1\Delta_2)$, and let x be an integer. Assume that at the initial step τ_o, there is no deviation cluster of size more than αx, and that at a step $\tau_1 \in J$, such that $\tau_1 > \tau_o$, there is a deviation cluster $Y^{(\tau_1)}$ of size $|Y^{(\tau_1)}| > x$. Then there is a step $\tau_\star \in J$ such that $\tau_o < \tau_\star < \tau_1$ for which there is a deviation cluster Z at step $\tau_\star$, satisfying that the size of Z is in the range $(\alpha x, x)$.

Proof. Assume through contradiction that for any turn $\tau \in (\tau_o, \tau_1)$, all the deviation clusters are of size smaller than αx. Particularly, at turn $\tau_1 - 1$, all the deviation clusters are of size smaller than αx. Yet, by Claim 1.11, the existence of the deviation cluster $Y^{(\tau_1)}$, at size d, implies the existence of a deviation cluster at turn $\tau_1 - 1$, of size at least αx, which leads to a contradiction. $\square$

Claim 1.13. Let C be a quantum circuit, with order J and depth d, and let $\mathcal{E} \sim \mathcal{N}_{\mathcal{E}}$, so the circuit $C[\mathcal{E}]$ is a random variable. For a time $t \leq 2d$ and integer $x \in \mathbb{N}$ let A be the event in which, until time t, there is a time $t' \leq t$ such that there is a deviation cluster of size greater than x. For any step $\tau \in J$, let B_τ be the event that step τ is the first step to have a deviation cluster of size in the range $(\alpha x, x)$. Then:

$$\mathbf{Pr}\left[A\right] \leq 2wd \max_{\tau \in [2wd]} \mathbf{Pr}\left[B_\tau\right]$$

Proof. By Corollary 1.12, we have that the occurrence of a divation cluster of size grether than x, before time time t implies that there is a step $\tau \in J$, such $\tau \leq 2wt$, in which there is a divation cluster of size greather than αx. Thus,

$$A \subset \bigcup_{\tau \in [2wt]} B_\tau \leq \bigcup_{\tau \in [2wd]} B_\tau$$

So by using the union bound we get:

$$\begin{aligned} \mathbf{Pr}\left[A\right] \leq \mathbf{Pr}\left[\bigcup_{\tau \in [2wd]} B_\tau\right] &\leq \sum_{\tau \in [2wd]} \mathbf{Pr}\left[B_\tau\right] \\ &\leq 2wd \max_{\tau \in [2wd]} \mathbf{Pr}\left[B_\tau\right] \end{aligned}$$

$\square$